An Internship Report

Of

Google AI-ML Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410031536

Name of Student: SATYAM KUMAR

Batch: 2024-2028

CANDIDATE'S DECLARATION\

I, SATYAM KUMAR, do hereby declare that the internship report titled "Electric Vehicle Modeling and Simulation Design Virtual Internship Program" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

All the references have been duly quoted, and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement. I further declare that I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: SATYAM KUMAR

Roll No.: 2410031536

ACKNOWLEDGEMENT

I am using this opportunity to express my sincere gratitude to everyone who supported and guided me throughout the **Electric Vehicle Modeling and Simulation Design Virtual Internship Program**. I am thankful for their valuable guidance, constructive feedback, and continuous encouragement during the completion of this internship. Their support and suggestions helped me enhance my knowledge and develop a better understanding of the concepts related to electric vehicle modeling and simulation.

I would like to express my sincere thanks to the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me with this valuable opportunity and a structured platform to learn and gain practical exposure in the domain of **Electric Vehicle Modeling and Simulation**. I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty members of the **School of Computer Science and Engineering** for their continuous support, motivation, and encouragement throughout the internship.

I would also like to express my heartfelt gratitude to my parents for their constant blessings, support, and encouragement, which motivated me to successfully complete this internship.

Signature:-

Student Name: SATYAM KUMAR

Roll No.: 2410031536

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शिक्षा मंत्रालय
MINISTRY OF EDUCATION

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

satyam kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Electric Vehicle Modeling and Simulation Design Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

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Chandan Pramanik
Director- Education
MathWorks India

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

INTERNSHIP GRADE O

Certificate ID : 4b1aac1408e940f8af9e

Student ID: STU699b1f6f2be871771773807

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter

Ref No: C17Y2026M081495075

Internship Offer Letter

Student Name	: satyam kumar
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Electric Vehicle Modeling and Simulation Virtual Internship Program
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: STU699b1f6f2be871771773807

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	MATLAB Onramp	Learn MATLAB®, including its language, development environment, and essential tools for analyzing science and engineering data.
Week 2	Simulink OnrampControl Design Onramp with Simulink	Simulink Onramp:Learn Simulink®, including creating, editing, and simulating models using block diagrams to represent real-world systems and algorithms.Control Design Onramp with Simulink:Learn feedback control design in Simulink®, including tuning PID controllers to achieve the desired closed-loop system performance.
Week 3	Simscape OnrampCircuit Simulation OnrampPower Electronics Simulation Onramp	Simscape Onramp:Learn Simscape® for modeling and simulating physical systems using a physical network approach without manually deriving equations.Circuit Simulation Onramp:Simulate analog electrical circuits in Simscape®, including modeling filters, power supplies, and analyzing their performance.Power Electronics Simulation Onramp:Simulate power electronics converters in Simscape™, including modeling and analyzing buck converters at different levels of fidelity.
Week 4	Stateflow Onramp	Learn Stateflow® for modeling supervisory control systems and decision logic using state machines and flow charts.
Week 5	Simscape Battery Onramp	Simulate Battery Management Systems (BMS) in Simscape®, including battery modeling, thermal management, and charging/discharging algorithms.
Week 6	Power Systems Simulation Onramp	Model and simulate electrical power systems by building a microgrid in Simulink® and Simscape Electrical™, while exploring power components and control strategies.
Week 7	System Composer Onramp	Learn model-based systems engineering using System Composer™, including system architecture, requirements tracing, and Simulink® integration for system verification.
Week 8	App Building Onramp	Build MATLAB® applications using App Designer, including user interface design, callbacks, and complete app development.

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the **Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program**, in the domain of **"Electric Vehicle Modeling and Simulation"**. The internship was carried out as part of the requirement for the **Bachelor of Technology (Computer Science and Engineering)** programme at **IILM University, Greater Noida**. The primary aim of this internship was to develop a strong practical foundation in **Electric Vehicle modeling, simulation, and related technologies**, and to gain hands-on exposure to the fundamental concepts involved in the design and analysis of electric vehicle systems.

The internship followed a structured learning approach combining **conceptual understanding with practical activities, assignments, simulations, and assessments**. Throughout the program, various concepts related to electric vehicle architecture, vehicle modeling, simulation techniques, electric powertrain systems, battery systems, and performance analysis were studied. The internship provided an opportunity to understand how modeling and simulation tools can be used to analyze and evaluate the performance of electric vehicle components and systems.

The successful completion of the internship provided valuable practical exposure and enhanced my understanding of **Electric Vehicle Modeling and Simulation**. The knowledge and skills gained during this internship will contribute to developing a strong technical foundation and understanding of modern electric mobility technologies.

4.2 Organization Profile

EduSkills Academy is an organization focused on enhancing the technical skills and employability of students through industry-oriented learning and virtual internship programs. EduSkills works in collaboration with the **All India Council for Technical Education (AICTE)** to provide students with opportunities to gain practical knowledge and industry exposure through the **AICTE – EduSkills Virtual Internship Program**.

The **Electric Vehicle Modeling and Simulation Design Virtual Internship Program** provides learners with structured learning and practical exposure to the fundamental concepts of electric vehicles, their modeling, simulation, and performance analysis. The program is designed to help students understand modern electric vehicle technologies and develop technical knowledge through guided learning activities, assignments, simulations, and assessments.

Through this internship program, students get an opportunity to connect theoretical concepts with practical applications in the field of **Electric Vehicle Modeling and Simulation**. The program contributes to developing technical skills and industry awareness, helping students build a foundation for further learning and professional development in emerging electric mobility technologies.

4.3 Problem Statement

Although electric vehicles are becoming an important part of modern transportation, many engineering students understand the concepts related to electric vehicle technology mainly at a theoretical level. There is often limited structured, hands-on exposure to electric vehicle modeling, simulation, system analysis, and performance evaluation using simulation-based approaches. Concepts such as electric vehicle architecture, battery systems, electric powertrains, vehicle dynamics, and simulation models can be difficult to understand without practical implementation and experimentation.

The problem addressed by this internship was to practically learn and apply the concepts of Electric Vehicle Modeling and Simulation through a structured learning and practical workflow. The internship focused on developing an understanding of electric vehicle components and their interactions, creating and analyzing simulation models, and evaluating important vehicle parameters and system performance. This practical approach helped bridge the gap between theoretical knowledge and the real-world application of electric vehicle technologies.

4.4 Project Objectives

- To understand the fundamental concepts and working principles of **Electric Vehicles (EVs)** and their major components.

- To study the basic architecture of electric vehicles, including the **electric powertrain, battery system, motor, controller, and energy storage system**.

- To develop an understanding of **electric vehicle modeling and simulation techniques**.

- To learn how to create and analyze simulation models for different **electric vehicle components and systems**.

- To study the performance characteristics of **electric motors and battery systems** used in electric vehicles.

- To analyze important electric vehicle parameters such as **speed, torque, power, energy consumption, and vehicle performance** through simulation.

- To understand the role of **battery management and energy management** in improving electric vehicle efficiency and performance.

- To gain practical exposure to the use of **simulation-based approaches** for analyzing and evaluating electric vehicle systems.

- To relate theoretical concepts of electric vehicles with their **practical modeling, simulation, and performance analysis**.

- To successfully complete the required **internship activities, assignments, assessments, and final evaluation** associated with the Electric Vehicle Modeling and Simulation Design Virtual Internship Program.

4.5 Scope of the Project

The scope of this internship covers the fundamental concepts and practical aspects of **Electric Vehicle Modeling and Simulation**, starting from understanding electric vehicle architecture and major components to developing and analyzing simulation models. The internship includes the study of important systems such as **electric motors, battery systems, power electronics, controllers, and electric powertrains**, along with their roles in the overall performance of an electric vehicle.

The project further covers the use of **modeling and simulation techniques** to study and evaluate electric vehicle performance. It includes analyzing parameters such as vehicle speed, torque, power, energy consumption, battery performance, and overall system efficiency through simulation-based activities. The internship also provides an understanding of how different vehicle components interact with each other and how simulation can be used to evaluate and improve system performance.

The scope of the internship was limited to a **virtual, simulation and learning-based environment** provided through the AICTE– EduSkills Virtual Internship Program. The work focused on conceptual learning, practical simulation activities, assignments, and assessments and did not involve the deployment or testing of an electric vehicle system in a live industrial or production environment.

4.6 Technologies and Tools Used

- **Modeling&Simulation:** Electric Vehicle modeling, system-level simulation, and performance analysis.
- **Electric Vehicle Systems:** Electric motors, battery systems, power electronics, controllers, and electric powertrain components.

- **Battery&Energy Analysis:** Battery performance analysis, energy consumption, charging and discharging characteristics.

- **Vehicle Performance:** Analysis of vehicle speed, torque, power, efficiency, and other important performance parameters through simulation.

- **Simulation-Based Learning:** Development, testing, and analysis of electric vehicle models in a virtual simulation environment.

- **Learning Platform:** AICTE– EduSkills Virtual Internship Program learning modules, practical activities, assignments, quizzes, and assessments.

4.7 System Architecture

The general architecture followed during the internship for the **Electric Vehicle Modeling and Simulation** learning track can be summarized in the following layers:

- **Vehicle Architecture Layer:** Defines the overall structure of the electric vehicle, including the battery, electric motor, power electronics, controller, and other major components.

- **Battery Layer:** Represents the battery energy-storage system and its important characteristics, including charging, discharging, voltage, current, and energy availability.

- **Powertrain Layer:** Models the interaction between the battery, power electronics, electric motor, and drivetrain to understand the flow of electrical and mechanical power.

- **Motor and Control Layer:** Represents the electric motor and its control system, focusing on the relationship between motor speed, torque, power, and vehicle operation.

- **Vehicle Dynamics Layer:** Represents the vehicle-level behavior by considering parameters such as vehicle speed, acceleration, torque requirements, power consumption, and driving conditions.
- **Simulation and Analysis Layer:** Combines the different models and allows the complete electric vehicle system to be simulated and analyzed. The simulation results are used to evaluate vehicle performance and energy efficiency.
- **Assessment Layer:** Includes the practical activities, assignments, quizzes, and assessments conducted during the internship to evaluate the learning and understanding of Electric Vehicle Modeling and Simulation concepts.

4.8 Methodology

The internship followed a structured, week-wise methodology combining **conceptual learning, practical simulation activities, hands-on exercises, and assessment-based evaluation**. The methodology was designed to provide a gradual understanding of **Electric Vehicle Modeling and Simulation**, beginning with MATLAB fundamentals and progressing through Simulink, Simscape, Stateflow, battery modeling, power systems simulation, system architecture, and application development. The overall learning process can be summarized as follows:

Week	**Module**	**Description**
Week 1	**MATLAB Onramp**	Learn MATLAB®, including its language, development environment, and essential tools for analyzing science and engineering data.
Week	**Simulink Onramp /**	Learn Simulink® for creating, editing, and

Week	Module	Description
2	**Control Design Onramp**	simulating models using block diagrams. Learn feedback control design and PID controller tuning.
Week 3	**Simscape Onramp / Circuit Simulation Onramp / Power Electronics Simulation Onramp**	Learn physical-system modeling using Simscape®, simulate analog electrical circuits, and model power electronic converters such as buck converters.
Week 4	**Stateflow Onramp**	Learn Stateflow® for modeling supervisory control systems and decision logic using state machines and flow charts.
Week 5	**Simscape Battery Onramp**	Simulate Battery Management Systems (BMS), including battery modeling, thermal management, and charging/discharging algorithms.
Week 6	**Power Systems Simulation Onramp**	Model and simulate electrical power systems by building a microgrid in Simulink® and Simscape Electrical™ and explore power components and control strategies.
Week 7	**System Composer Onramp**	Learn model-based systems engineering using System Composer™ , including system architecture, requirements tracing, and Simulink® integration.

Week	Module	Description
Week 8	**App Building Onramp**	Build MATLAB® applications using App Designer, including user-interface design, callbacks, and complete application development.

Note: There was no stipend associated with this internship. The internship followed a structured 8-week learning program consisting of MATLAB, Simulink, Simscape, Stateflow, Simscape Battery, Power Systems Simulation, System Composer, and App Building modules.

4.9 Expected Outcomes

For your **Electric Vehicle Modeling and Simulation Design Virtual Internship Program**, you can write this section as follows:

- **Ability to understand and explain** the fundamental concepts of Electric Vehicles (EVs), their architecture, and major components.

- **Practical understanding of electric vehicle modeling and simulation**, including the representation and analysis of different EV systems.

- **Hands-on exposure to battery modeling and analysis**, including charging, discharging, energy storage, voltage, current, and battery performance.

- **Ability to understand and analyze electric motor performance**, including speed, torque, and power characteristics.

- **Capability to model and evaluate an electric powertrain**, including the interaction between the battery, power electronics, motor, and drivetrain.

- **Skill to analyze electric vehicle performance** using simulation results, including speed, torque, power, energy consumption, and efficiency.

- **Ability to interpret simulation results** and understand the effect of different operating conditions on overall electric vehicle performance.

- **Successful completion of the required internship activities, assignments, practical simulations, and assessments**, demonstrating the knowledge and skills acquired during the internship.

4.10 Certificates of Completion and Communication Proof

The **Internship Offer Letter** and the **Internship Completion Certificate** issued by **EduSkills Academy**, confirming the successful completion of the **AICTE – EduSkills Virtual Internship Program in Electric Vehicle Modeling and Simulation**, are attached in **Chapter 3** of this report. The relevant official communication received during the internship, along with supporting evidence of the completed internship activities and assessments, is attached below as further proof of participation and successful completion of the internship program.

EduSkills Foundation 2 Jun

to me

Dear **satyam kumar**,

🎉 **Congratulations!**

You have successfully completed **10 weeks Siemens Data Science Master** internship with EduSkills Foundation. Your certificate is ready — please log in to your profile and download it.

📤 **Steps to Download Your Certificate:**

1. Login at: https://aicte-internship.eduskillsfoundation.org/
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your **LinkedIn profile** to get better job opportunities and visibility from corporates.

🌟 **Why You Should Share Your Achievements on LinkedIn**

1. **World's Largest Professional Platform**
   LinkedIn has **900+ million professionals and 60+ million companies** worldwide.
2. **Recruiters Actively Use LinkedIn**
   Around **8 out of 10 recruiters use LinkedIn** to find and verify candidates.
3. **Profiles Are Checked Before Hiring**
   More than **90% of hiring managers review LinkedIn profiles** before shortlisting candidates.

Reply Forward

6. BIBLIOGRAPHY/REFERENCES

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